

Problem #2

Due: Tuesday, October 28, turn in to canvas, as described in the syllabus.

1. Determine for which values of $\beta > 0$, there is “finite-time blow-up” of solutions (i.e. there exists a finite $T > 0$ such that $\lim_{t \rightarrow T} |x(t)| = +\infty$) for the following scalar ode:

$$\dot{x} = x^\beta, \quad x(0) = x_0 \in \mathbf{R}^+.$$

Specify the domain of existence of the solution, i.e. determine T as a function of x_0 and β .

In the remainder of this problem, let's focus on $\beta = 2$ and $x_0 = 1$. What is the time T to “blow up” in this case? Introduce the following transformation from t to a new time variable τ :

$$\tau = \int_0^t 1 + x^2(s) \, ds, \quad t \in [0, T)$$

where $x(s)$ satisfies the original problem $dx/ds = x^2$. Show that this is a one-to-one mapping between $t \in [0, T)$ and τ , and specify the range of τ . Let $y(\tau) = x(t(\tau))$; what differential equation does $y(\tau)$ satisfy? Does $y(\tau)$ have finite time blow-up? (You may find it helpful to read Section 4.3 of the textbook on Global Existence of Solutions.)

2. Consider the scalar initial value problem

$$\dot{x} = -|x|^\beta, \quad x(0) = 0.$$

For which values of $\beta > 0$ is the solution $x(t) = 0$ guaranteed to be unique on some interval $J = [-a, a]$, $a > 0$? If there is no such guarantee, explicitly construct a family of solutions to the initial value problem on the interval J .

3. Often in applications, the dynamics of a system being modeled, may be governed by quite different processes in different regions of the state space. And this might lead to a dynamical system that is set-wise defined, e.g.

$$\dot{\mathbf{x}} = \begin{cases} \mathbf{f}^+(\mathbf{x}), & \sigma(\mathbf{x}) > 0, \\ \mathbf{f}^-(\mathbf{x}), & \sigma(\mathbf{x}) < 0, \end{cases} \quad (1)$$

where the scalar equation $\sigma(\mathbf{x}) = 0$ defines the boundary between different regions of state space governed by different dynamics. (Note that I didn't say anything about the dynamics for points **in** the boundary.)

In the following, you explore how various possible definitions of the dynamics in the boundary might impact the behavior of the system. Specifically, consider (1) with $\mathbf{f}^\pm(\mathbf{x})$ given as a planar, linear vector field

$$\mathbf{f}^\pm(\mathbf{x}) = \begin{pmatrix} -1 \pm x_2 \\ x_2 \mp x_1 \end{pmatrix},$$

and let the boundary be defined by $\sigma(\mathbf{x}) = x_2$.

- (a) Plot the phase portrait associated with $\dot{\mathbf{x}} = \mathbf{f}^+(\mathbf{x})$ and $\dot{\mathbf{x}} = \mathbf{f}^-(\mathbf{x})$, separately in the square region $x_1, x_2 \in [-2, 2]$, i.e. don't restrict them to their regions $x_2 > 0$ or $x_2 < 0$.
- (b) Show that the vector field is discontinuous on $x_2 = 0$.

- (c) Let's proceed by restricting to consideration of the forward time behavior of the model with some prescribed dynamics on the discontinuity boundary. For example, one natural way that the dynamics might be defined is as some convex combination of $\mathbf{f}^+$ and $\mathbf{f}^-$, i.e. let

$$\dot{\mathbf{x}} = \alpha \mathbf{f}^+(\mathbf{x}) + (1 - \alpha) \mathbf{f}^-(\mathbf{x}), \text{ for } \mathbf{x} \text{ satisfying } \sigma(\mathbf{x}) = 0,$$

where $\alpha \in [0, 1]$ is a parameter. For the following choices of α , plot the forward solution in the phase plane for an initial condition $(x_1, x_2) = (2, 1)$. (This can be done as a somewhat qualitative sketch.)

- (i) $\alpha = 0$, (ii) $\alpha = 0.5$, (iii) $\alpha = 1$. Comment on whether the behavior is sensitive to the choice of the parameter $\alpha \in [0, 1]$.

4. Use Hartman-Grobman theorem to prove that linear asymptotic stability implies asymptotic stability for an equilibrium $\mathbf{x}^*$ of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$, where $\mathbf{f} : \mathbf{R}^n \rightarrow \mathbf{R}^n$ is a C^1 vector field. Can you also use it to show that linear instability implies instability?
5. In Section 4.8 of the textbook, on the Hartman Grobman theorem, there is an example of constructing a homeomorphism $H(x, y) = (x, y - x^2/3)$ that satisfies the topological conjugacy condition $\psi_t \circ H = H \circ \phi_t$ where ϕ_t is generated by

$$\begin{aligned}\dot{x} &= x \\ \dot{y} &= -y + x^2.\end{aligned}$$

Solve these differential equations to obtain ϕ_t and ψ_t , where ψ_t applies to the linearized flow. Then check, by explicit computation, that indeed $\psi_t \circ H = H \circ \phi_t$.

6. Consider the Lorenz equations

$$\begin{aligned}\dot{x} &= \sigma(y - x) \\ \dot{y} &= rx - y - xz \\ \dot{z} &= xy - bz,\end{aligned}$$

where r, b, σ are positive constants and $(x, y, z) \in \mathbb{R}^3$.

- (a) Show that the origin is the only fixed point for $r \in (0, 1]$, and that the following is a strong Lyapunov function for it for $r < 1$:

$$L = \frac{1}{2} \left(\frac{x^2}{\sigma} + y^2 + z^2 \right).$$

- (b) Show that L is only a weak Lyapunov function if $r = 1$, but that you can use LaSalle's Invariance Principle to prove that the origin is asymptotically stable for $r = 1$ as well as for $r < 1$. Note that you did not have to restrict any of this to a particular neighborhood U of the origin. The origin is globally attracting for the Lorenz equations when $r \in (0, 1]$.